

README: gTREND-Nitrogen Dataset

(Version 1.0 – March 2026)

1. Overview

gTREND-Nitrogen is a 250-m grid scale dataset of annual nitrogen (N) fluxes and surplus for the contiguous United States covering the period from 1930–2017 (Fig. 1). In developing this dataset, we have combined county-scale data from the TREND-Nitrogen dataset with gridded land use and population data to provide fine-scale estimates of major N inputs (fertilizer, manure, biological nitrogen fixation, human waste, atmospheric deposition) and outputs (crop and pasture uptake), allowing for calculation of annual nitrogen surplus.

Key Dataset Characteristics: gTREND-Nitrogen

- Temporal coverage: 1930–2017
- Spatial Coverage: Contiguous U.S. (CONUS)
- Spatial Resolution: 250 m grid
- Units $\text{kg-N ha}^{-1} \text{ yr}^{-1}$
- Coordinate Reference System: EPSG:5070

For additional discussion and presentation of the datasets, see:

Chang, S.Y., Byrnes, D.K., Basu, N.B. & Van Meter, K.J. (2026). *gTREND-Nitrogen – Long-term nitrogen mass balance data for the contiguous United States (1930–2017)*. Scientific Data. <https://doi.org/10.1038/s41597-026-06576-x>

2. Authors

S.Y. Chang^{1*} - shuyu.chang.hydro@gmail.com

D. K. Byrnes^{3*} - danyka.byrnes@proton.me

N.B. Basu^{3,4,5} - nandita.basu@uwaterloo.ca

K.J. Van Meter^{1,2} - vanmeterlab@gmail.com

*Shared first-authorship

3. Affiliations

1. Department of Geography, The Pennsylvania State University, State College, PA
2. Earth and Environmental Systems Institute, The Pennsylvania State University, State College, PA
3. Department of Civil and Environmental Engineering, University of Waterloo, Waterloo, Ontario, Canada

4. Department of Earth and Environmental Sciences, University of Waterloo, Waterloo, Ontario, Canada
5. The Water Institute, University of Waterloo, Waterloo, Ontario, Canada

4. Dataset Contents

- Gridded data (gTREND-Nitrogen): 22 nitrogen mass balance components downscaled to 250-m resolution (1930-2017) in GeoTIFF format
- Aggregated N surplus estimates at hydrologic unit code 8 (HUC8) level (1930-2017)

4.1 *gTREND-Nitrogen*

The gTREND-Nitrogen dataset covers the contiguous U.S. and includes 22 Nitrogen mass balance components for the years 1930-2017. The gTREND-Nitrogen data is provided as zipped subfolders and organized around different components of the N mass balance. Within the zipped folders, data is available in the Tag Image File Format (TIFF) with units of $\text{kg-N ha}^{-1}\text{y}^{-1}$ at 250-m grid scale. Each TIFF file represents a different annual N mass balance component (see **Table 1**). In the file names, the YYYY values refer to the data year.

We also provide aggregated N surplus estimates at hydrologic unit code 8 (HUC8) level from 1930 to 2017 in the zipped HUC_scale folder, which contains one csv file (Nsurplus_HUC8.csv) and one shapefile (HUC8.gpkg). The HUC8.shp contains the HUC8 boundaries for the continental United States, which is used to extract basin-averaged N surplus values. In Nsurplus_HUC8.csv, the first column is named “HUC8,” which contains the unique HUC8 basin identifiers. The csv also includes one column for each year of the dataset, with the naming convention “YYYY_Nsurplus,” with YYYY values referring to the data year. The data in these columns corresponds to aggregated basin-averaged N surplus values with units of $\text{kg-N ha}^{-1}\text{y}^{-1}$ from 1930 to 2017.

Table 1. Folder and file structure of the gTREND-Nitrogen dataset in the data repository.

Zipped Subfolder Name	File name	Zipped Folder Content	
		File Naming Convention	Number of Files
Oxidized Atmospheric N Deposition	Atmospheric_Oxidized.zip	Atmospheric_Oxidized_YYYY.tif	88
Reduced Atmospheric N Deposition	Atmospheric_Reduced.zip	Atmospheric_Reduced_YYYY.tif	88
Crop and Pasture N Uptake	Agriculture_Uptake.zip	CropUptake_Ag_YYYY.tif	88
Crop N Uptake	Cropland_Uptake.zip	CropUptake_Cropland_YYYY.tif	88
Pasture N Uptake	Pasture_Uptake.zip	CropUptake_Pasture_YYYY.tif	88
Farm Fertilizer	Agriculture_Fertilizer.zip	Fertilizer_Ag_YYYY.tif	88
non-Farm Fertilizer	Domestic_Fertilizer.zip	Fertilizer_Dev_YYYY.tif	88
Biological N Fixation (all-agriculture)	Agriculture_Fixation.zip	Fix_Ag_YYYY.tif	88
Cropland N Fixation	Cropland_Fixation.zip	Fix_Cropland_YYYY.tif	88
Pasture N Fixation	Pasture_Fixation.zip	Fix_Pasture_YYYY.tif	88
Human Waste N	Human.zip	Human_YYYY.tif	88
Livestock Waste N - Milk Cows	Lvst_MilkCows.zip	MilkCows_YYYY.tif	88
Livestock Waste N - Beef Cows	Lvst_BeefCows.zip	BeefCows_YYYY.tif	88
Livestock Waste N - Other Cattle	Lvst_OtherCattle.zip	OtherCattle_YYYY.tif	88
Livestock Waste N - Horses	Lvst_Equine.zip	Equine_YYYY.tif	88
Livestock Waste N - Hogs	Lvst_Hogs.zip	Hogs_YYYY.tif	88
Livestock Waste N - Layers & Pullets	Lvst_LayersPullets.zip	LayersPullets_YYYY.tif	88
Livestock Waste N - Broilers	Lvst_Broilers.zip	Broilers_YYYY.tif	88
Livestock Waste N - Turkeys	Lvst_Turkeys.zip	Turkeys_YYYY.tif	88
Livestock Waste N - Sheep & Goats	Lvst_SheepGoat.zip	SheepGoat_YYYY.tif	88
Livestock Waste N - All	Lvst_Sum.zip	Livestock_YYYY.tif	88
N Surplus	Surplus.zip	Surplus_N_YYYY.tif	88
HUC8-scale N surplus magnitudes	HUC8_scale.zip	Nsurplus_HUC8.csv HUC8.gpkg	2

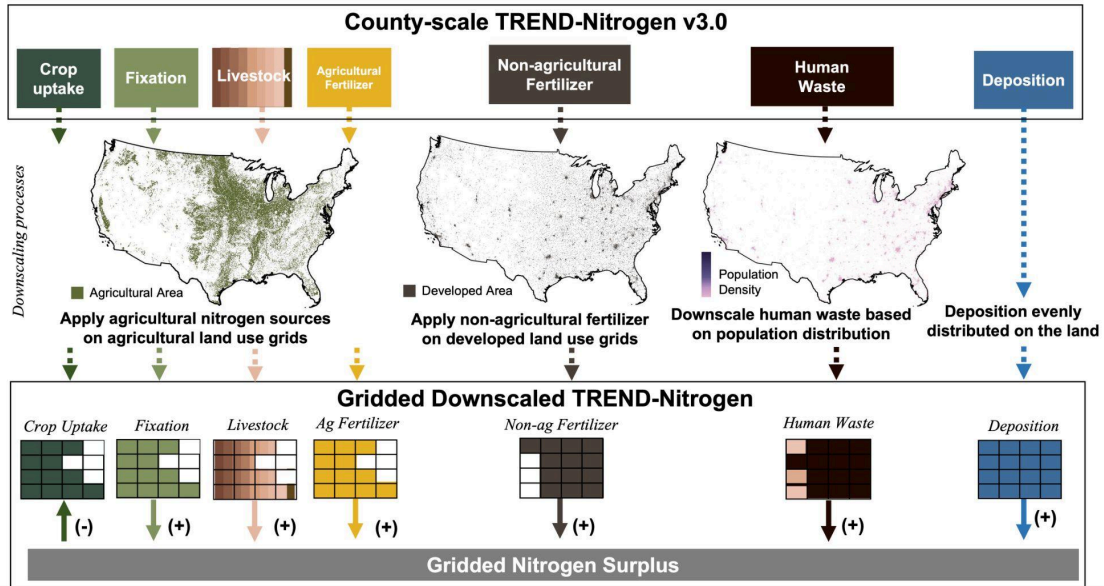


Figure 1. Data sources and overall workflow for creating the downscaled gTREND-Nitrogen dataset.

5. Methodology

County-scale N fluxes from TREND-Nitrogen were downscaled to the 250-m grid scale according to land use/land cover and population densities, as described below. For the downscaling, N fluxes were divided into three different land-use categories: (1) agricultural, (2) developed/urban, and (3) all land (**Fig. 2**). Agricultural N fluxes were considered to include four major landscape flux categories: (1) N fertilizer, (2) N in livestock manure N, (3) biological N fixation in cropland and pastureland, and (4) N uptake by crops and pasture. Developed/urban fluxes were considered to include (1) N in human waste and (2) non-farm N fertilizer. Finally, atmospheric N deposition was considered to be an all-land flux.

Land-use trajectories were established based on two different land use and land cover (LULC) datasets: (1) a modeled historical LULC dataset created at a resolution of 250 m (hereafter referred to as the Sohl LULC dataset)^{1,2}, and (2) the widely used National Land Cover Database (NLCD), available at a resolution of 30 m.³ These two datasets were integrated and harmonized to reconstruct temporally continuous and spatially explicit land-use trajectories from 1930-2017 at the 250-m grid scale for the contiguous United States (CONUS). Specifically, we use here the annual 250-m Sohl LULC data from 1938 to 2005, and then use 30-m NLCD data for the years 2006, 2008, 2011, 2013, 2016. The NLCD data was resampled from 30 m to 250 m using the QGIS-python Nearest Neighbor Function, version 3.28.⁴ Consistent land use was assumed for the years 1930-1937 and in the years between the years of NLCD availability after 2005.

Gridded population-density data was obtained from the 1-km gridded, decadal population maps developed for the contiguous U.S. by Fang & Jawitz.^{5,6} These maps were created using a parsimonious modeling approach, in which decennial U.S. census data was brought together with a variety of other factors, including socioeconomic desirability, suitability, and inhabitability. While five different population models of varying complexity were created in the Fang & Jawitz study,⁵ we use here the results of the M3 model, which was of intermediate complexity but found to be the most accurate based on validation using county subdivision population data. To provide annual estimates of population for the gTREND-Nitrogen downscaling work, we used linear interpolation between the modeled years, and then assumed a constant population for the years 2011-2017. The annual 1-km maps of human population were resampled to a 250-m resolution using the QGIS-python Nearest Neighbor Function, version 3.28 to align with other gridded components of the gTREND-Nitrogen dataset.

5.1 Agricultural N Fluxes

To prepare for downscaling of the agricultural N fluxes, annual, gridded binary land-use maps were created to classify 250-m grid cells as either agricultural or non-agricultural, $gLU_{ag}(k, t) = 0/1$, where k refers to the individual grid cell and t to year. Agricultural land use is defined here as areas used for the production of crops, areas being actively tilled, and areas of grasses, legumes, or grass-legume mixtures planted for livestock grazing or the production of seed or hay. Specifically, note that agricultural land was considered to include both cultivated cropland and hay/pasture in both the Sohl LULC dataset and the NLCD. Land not falling into either the developed or agricultural land classes was considered to be non-agricultural land, $gLU_{ag}(k, t) = 0$.

Total agricultural area in each county i in each year t , $AREA_{ag}(i, t)$ (ha), was then calculated as the sum of the gridded agricultural land. For each agricultural component of the N mass balance, j , county-scale N flux values normalized by agricultural area, $N_{ag, cnty}(i, j, t)$ ($\text{kg ha-agland}^{-1} \text{y}^{-1}$) were calculated by dividing the total county N flux by the time-varying agricultural area of each county, i . County-scale vector maps of $N_{ag, cnty}$ values were then rasterized to a 250-m grid scale. To appropriately assign agricultural N fluxes only to grid cells with agricultural land use, the county-scale flux values were multiplied by the 0 or 1 values in the binary land-use grid.

5.2 N Fluxes on Developed Land

As noted above, N fluxes on developed land were considered to include non-farm N fertilizer and human waste. For the non-farm fertilizer, the county-scale estimates of non-farm fertilizer, $FERT_{nf}$ (kg-N y^{-1}), were downscaled based on developed land use, assuming that homes and businesses using fertilizer would not occur on either agricultural or “other” land. Using a method similar to that developed for the agricultural

N fluxes, we created binary maps to classify 250-m grid cells as either developed or non-developed, $gLU_{dev}(k,t) = 0/1$, where k refers to the individual grid cell and t to year. Land not falling into either the developed or agricultural land classes was considered to be non-developed, $gLU_{dev}(k,t) = 0$.

Total developed area in each county in each year, $AREA_{dev}(i,t)$ (ha), was then calculated as the sum of the gridded developed land in that county in year t . County-scale vector maps of $FERT_{nf,dev}$ values were then rasterized to a 250-m grid scale. Next, to assign the non-farm N fertilizer fluxes only to grid cells with developed land use, the county-scale flux values were multiplied by the 0 or 1 values in the binary land-use grid.

For human waste, annual, grid-scale estimates of N from human waste, $gHUM(k,t)$ ($kg-N ha^{-1} y^{-1}$) were developed using gridded estimates of human population⁵ and literature-based estimates of per capita annual human N excretion ($5 kg-N pp^{-1} y^{-1}$).⁷

5.3 All-Land N Fluxes (Atmospheric Deposition)

The county-scale oxidized and reduced atmospheric N deposition magnitudes, $DEP_{ox}(i,t)$ and $DEP_{red}(i,t)$, were normalized by county area, $AREA(i,t)$:

$$DEP_{ox,norm}(i,t) = \frac{DEP_{ox}(i,t)}{AREA_{cnty}(i,t)} \quad (\text{Eq. 1})$$

$$DEP_{red,norm}(i,t) = \frac{DEP_{red}(i,t)}{AREA_{cnty}(i,t)} \quad (\text{Eq. 2})$$

For the all-land fluxes, all grid cells of our 250-m gridded map of the U.S. were assigned 1 values, $gLU_{all}(k,t) = 1$, where k refers to the individual grid cell and t to year. The normalized deposition flux values were then multiplied by the 1 values in the land-use grid.

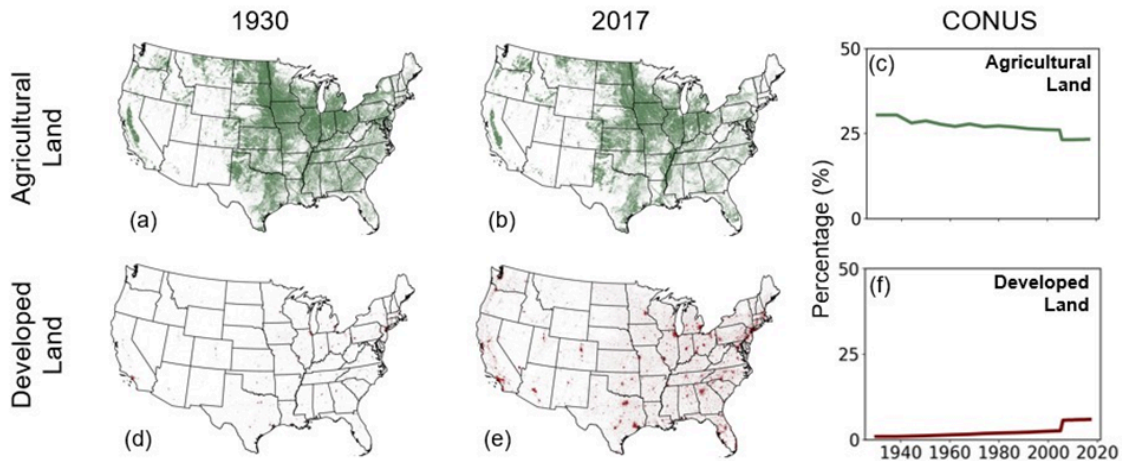


Figure 2. Agricultural and developed land use within the contiguous U.S (a,b,d,e). Land-use categories are based on a simplification of categories in the NLCD and Sohl LULC datasets. Land-use trajectories for each land category are shown in (c) and (f).

5.4 Gridded N Surplus

The annual, grid-scale estimates of the N surplus, gNS , were calculated by raster summation of the gridded components of the N mass balance. Here, we define the N surplus as a sum of N inputs (fertilizer N, biological N fixation, atmospheric N deposition, manure N, and human N) and non-hydrologic N outputs (crop and pasture N uptake):⁸

$$gNS(k, t) = gFERT(k, t) + gBNF(k, t) + gDEP_{ox}(k, t) + \quad (Eq. 3)$$

$$+ gMAN(k, t) + gHUM(k, t) - gCROP(k, t) - gPAST(k, t)$$

where $gFERT$ is the sum of farm and non-farm N fertilizer, $gBNF$ is biological N fixation, $gDEP$ is oxidized N deposition (both wet and dry forms), $gMAN$ is N in livestock manure, $gHUM$ is the N in human waste, $gCROP$ is N uptake in crops, and $gPAST$ is N uptake by pasture grasses consumed by livestock, all in units of $\text{kg ha}^{-1}\text{yr}^{-1}$.

6. Known limitations

While the dataset is internally consistent, several limitations arise from the data sources used and methodological assumptions.

6.1 Temporal Continuity

As noted in the Methods, land-use trajectories are derived from two harmonized datasets (Sohl LULC and NLCD). Minor discontinuities appear across the 2005-transition due to differences in classification schemes and resolution, but these do not reflect real-world abrupt changes in land use.

Population data prior to 1990 rely on linear interpolation between decadal census counts, introducing uncertainty for intermediate years.

6.2 Downscaling assumptions

County-scale nitrogen fluxes were allocated using binary land-use masks and population density grids. This approach maintains spatial fidelity but cannot capture within-class heterogeneity, e.g. variations in fertilizer rates within agricultural areas.

6.3 Simplified representations of historical practices

Historical livestock weights, and crop nitrogen concentrations are based on literature values and interpolated across time.

Wastewater treatment and manure management practices are not explicitly modeled. As such, human and livestock nitrogen inputs are treated within the mass balance as fully retained within the landscape.

6.4 Suitability for use

Overall, the dataset is best suited to watershed- or regional-scale analyses of long-term nitrogen trends and nutrient balances rather than field-level management decisions. Future updates may refine land-use harmonization, enhance downscaling approaches, and incorporate more explicit representation of waste treatment pathways.

7. Versioning and citation

7.1 Version

Current release: v1.0 (March 2026)

This version represents the first public release of the gTREND-Nitrogen dataset, covering 1930-2017.

Future updates will focus on improved land-use harmonization and refinement of downscaling methodologies.

7.2 How to cite

All components of the dataset share the same DOI. Please use the following formats

Chang, S.Y., Byrnes, D.K., Basu, N.B., & Van Meter, K.J. (2026). “gTREND-Nitrogen - Long-term nitrogen mass balance data for the contiguous United States (1930-2017).” Version 1.0. Figshare [<https://doi.org/10.6084/m9.figshare.29897375>]

7.3 Version history

- **V1.0** - Initial public release (March 2026)

8. Funding and acknowledgments

This research was supported in part by a NASA NIP grant to KVM under Grant #80NSSC22K0542, a USDA NIFA grant to KVM under grant number UDF0000221, and startup funding from The Pennsylvania State University. The research was also supported through NASA FINESST funding to KVM and SYC under Grant #80NSSC22K1440. We

also acknowledge funding support for NBB and DKB from the Lake Futures Grant, which is a part of the Global Water Futures Project, funded through the Canada First Research Excellence Fund, and from the Canada Research Chair Program. Additional funding support was provided to DKB by the NSERC Alexander Graham Bell Canada Graduate Scholarship.

Special thanks to collaborators and reviewers who provided feedback on early versions of this dataset, ensuring its accuracy and usability. We would also like to thank the broader research community whose previous work on nitrogen and phosphorus fluxes provided methodological foundations for this dataset.

9. Companion Datasets

TREND-Phosphorus and gTREND-Phosphorus (v1.0) - TREND-Phosphorus and gTREND-Phosphorus. Figshare+. Dataset.
[<https://doi.org/10.25452/figshare.plus.30158131>]

10. Code availability

All code used to downscale county-level data to the 250-m grid for generating gTREND-Nitrogen is available on the project's GitHub repository (<https://github.com/blue-node-hydrologics/gTREND-Nitrogen>). The repository also includes an aggregation script that allows users to summarize the gridded data by watershed or by administrative boundaries.

11. Contact and future plans

For questions, feedback, or to report issues with the dataset, please contact:

- Shuyu Chang – shuyu.chang.hydro@gmail.com
- Danyka Byrnes – danyka.byrnes@proton.me
- Nandita Basu – nandita.basu@uwaterloo.ca
- Kim Van Meter – vanmeterlab@gmail.com

We welcome collaboration and feedback to improve future versions of gTREND-Nitrogen.

12. References

1. Sohl, T. *et al.* Modeled Historical Land Use and Land Cover for the Conterminous United States: 1938-1992. (2016).
2. Sohl, T. L. *et al.* Spatially explicit modeling of 1992-2100 land cover and forest stand age for the conterminous United States. *Ecol. Appl.* **24**, 1015–1036 (2014).
3. Conterminous United States Land Cover Projections - 1992 to 2100.
<https://www.sciencebase.gov/catalog/item/5b96c2f9e4b0702d0e826f6d> (2021).
4. QGIS. <https://www.qgis.org/en/site/>.
5. Fang, Y. & Jawitz, J. W. High-resolution reconstruction of the United States human population distribution, 1790 to 2010. *Sci Data* **5**, 180067 (2018).
6. Fang, Y. & Jawitz, J. W. The evolution of human population distance to water in the USA from 1790 to 2010. *Nat. Commun.* **10**, 430 (2019).
7. Hong, B., Swaney, D. P. & Howarth, R. W. A toolbox for calculating net anthropogenic nitrogen inputs (NANI). *Environ. Modell. Softw.* **26**, 623–633 (2011).
8. Byrnes, D. K., Van Meter, K. J. & Basu, N. B. Long-term shifts in U.S. nitrogen sources and sinks revealed by the new TREND-nitrogen data set (1930–2017). *Global Biogeochem. Cycles* **34**, (2020).